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Rezki Boureghdad

Software Engineer · Back-end

PROFILE

Software engineer specialising in back-end, and a founding member of Solrus, a software studio in Algiers. I build the Python services, PostgreSQL data models and REST APIs behind large, multi-service B2B platforms (FastAPI, Kafka, Redis), and I find and fix what slows them down: login cut from 6.2 s to 0.2 s, catalogue reads from 420 ms to 18 ms. Freelance full-stack developer since 2021, and a part-time computer-science teaching assistant.

EXPERIENCE

Data Specialist, Structure & Architecture (Back-end Engineer) · Founding Member

Jun 2025 – Present

Solrus — software studio, team of 4–8 · Algiers, remote-first

- Back-end and data engineering for large, multi-service B2B platforms (confidential clients): domain models, PostgreSQL schemas and Alembic migrations, REST API contracts, third-party integrations.
- Build distributed, event-driven Python services (FastAPI, SQLAlchemy 2, Kafka, Redis, Celery).
- Found and fixed a bottleneck in an event-driven auth flow: login time from 6.2 s to 0.2 s.
- Load-tested a platform with k6 at 200 concurrent users, then fixed what it exposed: password hashing off the event loop (p95 −18% overall, −31% on the CRM) and paginated catalogue reads (420 ms → 18 ms per page).
- Sped up a matching engine with a teammate (typical run 94 → 28 ms), and cut the test suite from 290 s to 44 s with parallel pytest and per-worker databases.
- Security by design: role-based access, TOTP 2FA, audit trails, data residency. Ran an internal grey-box review (about 20 checks) before an external pen test: no serious vulnerability, 4 low findings fixed.
- Write test scenarios for teammates' features (pytest on real PostgreSQL, Playwright end to end) and send findings back before release, on an API test suite that grew from about 200 to 1,600+ tests.
- QA and debugging for **Zakat.business**, the founding team's non-profit platform (Label award, 20+ companies).

Freelance Full-Stack Developer

May 2021 – Present

Self-employed (auto-entrepreneur) · contract basis

- 10+ client projects and feature builds (web, mobile, AI), from scoping to deployment, including e-motion, Ahl Quran and 7alkaty (see projects).

Teaching Assistant — Computer Science

Nov 2024 – Present

Université M'Hamed Bougara, Boumerdès · part-time, up to 12 h/week

- Tutorials and labs for up to 8 groups of 30–40 students across 12 subjects (algorithms, databases, language theory, compilation), with a 70% pass rate.

EDUCATION

MSc Software Engineering & Data Science
Université M'Hamed Bougara, Boumerdès · graduated Jan 2024

BSc Computer Science
Université M'Hamed Bougara, Boumerdès · graduated Jan 2022

CERTIFICATIONS

NVIDIA
Fundamentals of Deep Learning · Intro to Machine Learning

Neo4j
Graph Data Science · Fundamentals · Graph Data Modeling

EF SET English
73/100 — C2 Proficient

LANGUAGES

Arabic
Native

English
C2 (EF SET)

French
Professional working

SELECTED PROJECTS

claim-bot — migration-number Discord bot

Own tool

- Reserves the next migration number before the file exists, so developers never collide. Merges committed files from the GitHub API with open claims, and checks the Alembic chain for extra heads, broken links and cycles.

Python · discord.py · GitHub GraphQL · Alembic

DocMCP — local PDF search server for AI tools

Open source

- Ingests PDF folders into ChromaDB and serves them to any MCP client (Kiro, Claude Code) as semantic-search tools. Fully local: ONNX / sentence-transformer embeddings, LLM on Ollama.
- Recursive Language Model reasoning: a local LLM explores the full documents in a Python REPL instead of relying on top-k chunks. Web UI and one-command Docker setup.

Python · Node.js · MCP · ChromaDB · Ollama · Docker · github.com/V0idSeeker/RAG-MCP

Forsa / Aamal ATS — recruitment platform

Solrus public case study

- Connects companies, independent recruiters and job seekers in Algeria. Semantic job matching with sentence-transformer embeddings and location search with PostGIS.
- WebSocket chat, community forum, FCM push notifications, SlickPay subscriptions, JWT/Argon2 auth. Flutter mobile app and admin panel.

FastAPI · PostgreSQL/PostGIS · Flutter · Docker

e-motion — back-office for cash-on-delivery shops

Freelance · in use

- Built for an e-commerce business that runs its orders, stock, couriers and expenses on it, with real profit per order. Weighted-average purchase cost, stock that can never go below zero, alerts for customers who often cancel or return, and an audit log.
- Excel import, packing slips, English/French interface.

Next.js · TypeScript · Supabase · Vercel

VoidSketches — free CS lessons through analogies

Own product

- Animated, narrated lessons in English and French, one concept each, starting with compilation. Local TTS pipeline whose timings drive animation and captions.

React · Vite · TTS · voidsketches.vercel.app

Community apps — Ahl Quran & 7alkaty

Freelance

- **Ahl Quran:** multi-tenant Quran-school manager for study circles, attendance and progress (FastAPI, PostgreSQL, Flutter).
- **7alkaty:** mosque course and imam-following app with scheduled push notifications (Node/Express, MongoDB, Expo).

Forest-fire detection & reporting

Master's thesis

- Co-designed a three-part system: a citizen reporting app, a data platform and alerting. Detection models built with YOLO and Faster R-CNN.

Python · YOLO · R-CNN · Flutter · Firebase

SKILLS

Back-end
Python, FastAPI, Django/DRF, SQLAlchemy 2 (async), Alembic, Pydantic, Celery, REST APIs, Node/Express

Data
PostgreSQL/PostGIS, SQL, Redis, Kafka, MongoDB, Neo4j, Supabase

AI / ML
RAG, MCP servers, FAISS/Chroma, embeddings & reranking, LLM APIs, YOLO/R-CNN

Front & mobile
TypeScript, Next.js, React, Flutter/Dart

Ops & quality
Docker, nginx, Linux servers (SSH, xrdp), OVH, VPS hosting, GitHub Actions, Vercel, Grafana/Loki, pytest, Playwright

Security
JWT, Argon2, OAuth2, TOTP 2FA, ISO 27001, data residency

Also
Contract-first API design, decision records, test scenarios as review, AI-assisted development

CS FOUNDATIONS

Algorithms & complexity

Compilers & formal languages

Formal methods

Deep learning

Programming-language semantics

Database theory & graph databases